

The Progressive Overload Training Model: Physiological Mechanics, Biomechanical Constraints, and Algorithmic Program Synthesis for Female Populations

The progressive overload training model serves as the foundational architecture of modern resistance training, predicated on the systematic and gradual increase of physical stress to elicit favorable biological adaptations in strength, hypertrophy, and functional capacity.¹ At its core, the model identifies that the human body exists in a state of dynamic equilibrium, or homeostasis, and that any persistent improvement in physical performance requires a calculated disruption of this state through escalating stressors.⁴ To develop a robust resource for an automated programming environment, it is necessary to synthesize the classical physiological theories of adaptation with contemporary biomechanical constraints and sex-specific biological variables.

Foundations of Physiological Adaptation: GAS and SRA Theory

The mechanical success of progressive overload is rooted in the General Adaptation Syndrome (GAS), a tripartite response to stress first conceptualized by Hans Selye and later refined for sports science by practitioners like John Garhammer.⁷ When the musculoskeletal system is exposed to a novel or more intense training stress—such as a heavier weight, increased volume-load, or shorter rest intervals—it enters the Alarm Phase.⁶ This initial period is characterized by an accumulation of fatigue, soreness, stiffness, and a temporary reduction in performance capacity as the body's energetic stores are depleted.⁸

Following the alarm phase, provided that the stressor is not catastrophic and recovery is prioritized, the system transitions into the Resistance Phase.⁸ During this window, the organism implements specific biochemical, structural, and mechanical adjustments—such as myofibrillar protein synthesis and improved neural drive—to return to a state of functional capacity that exceeds the previous baseline.⁶ This phenomenon, termed supercompensation, represents the "overshooting" of the original homeostatic level.⁵ If the training stimulus is repeated too frequently without adequate recovery, or if the load is increased too aggressively, the system enters the Exhaustion Phase, leading to non-functional overreaching or overtraining syndrome,

where adaptive reserves are depleted and performance collapses.⁸

The timing of the next training stimulus is critical to the long-term success of the overload model. The Stimulus-Fatigue-Recovery-Adaptation (SRA) theory, an extension of GAS, suggests that the magnitude of the workload determines the duration of the recovery curve.⁸ High-intensity sessions targeting large muscle groups require longer windows for fatigue to dissipate before supercompensation is realized.⁸ Conversely, if the subsequent session occurs too late, the athlete enters a state of involution or detraining, where the gains from the previous cycle are lost as the body returns to its original baseline.⁶ Therefore, progressive overload is not merely about "lifting more," but about the precise synchronization of stress and recovery to ensure the "wave" of fitness trends upward over time.⁵

Phase of Adaptation	Physical Manifestation	Biological Mechanism
Alarm (Shock)	Acute fatigue, muscle soreness, performance drop ⁶	Glycogen depletion, cortisol release, calcium release impairment ⁷
Resistance (Adaptation)	Increased strength, improved recovery speed ⁸	Myofibrillar hypertrophy, increased capillarization, enzymatic activity ⁵
Supercompensation	Performance exceeds previous baseline ⁵	Structural reinforcement of connective tissue and muscle fibers ⁷
Exhaustion (Overtraining)	Chronic fatigue, persistent pain, psychological burnout ⁸	CNS down-regulation, systemic inflammation, neurohormonal imbalance ¹⁰

Mechanical Variables of the Overload Model

The implementation of progressive overload requires the manipulation of several distinct variables, or "levers," to maintain a constant state of adaptation.² In an AI-driven programming context, these variables must be adjusted based on the specific phase of training and the user's individual response data.

Absolute Load and Intensity

Increasing the weight lifted is the most recognized form of progressive overload, primarily targeting high-threshold motor unit recruitment and mechanical tension.² If a user can

complete a given weight for a prescribed number of repetitions with perfect technique, a small increase (e.g., 2.5–5 lbs or 5–10%) provides a novel stimulus that forces the nervous system to adapt to higher force requirements.² This method is especially effective for strength-focused phases of training.²

Volume: Sets, Repetitions, and Tonnage

Training volume, often calculated as $Volume = Sets \times Reps \times Weight$, is a primary driver of muscle hypertrophy and metabolic work capacity.² Increasing volume can involve adding an extra set per exercise, increasing the number of repetitions within a set at the same load, or increasing the total weekly workload per muscle group.² For muscle growth, an initial target of 8–15 sets per muscle group per week is often recommended, with higher priorities (e.g., specific weak points) receiving up to 15 sets.¹²

Training Frequency and Density

Frequency refers to how often a muscle group is trained within a week.² While high frequency (3–5 days per week) can increase total weekly volume and provide more frequent protein synthesis triggers, it requires careful management of recovery to avoid overtraining.² Density, on the other hand, involves increasing the amount of work done in a specific timeframe by reducing rest intervals.² For instance, reducing rest from 180 seconds to 60–90 seconds increases metabolic stress and cardiovascular demand, improving muscular endurance even if the load remains constant.²

Temporal and Complexity Variables

Time Under Tension (TUT) involves altering the tempo of a lift—specifically slowing the eccentric (lowering) phase or adding pauses at the bottom—to increase the ischemic stress and metabolic byproducts within the muscle.² This provides a novel stimulus for hypertrophy and movement control without the need for heavier absolute loads.² Finally, exercise complexity allows for overload through movement progression, such as transitioning from a TRX squat to a kettlebell goblet squat and eventually a barbell back squat.¹²

The Squat University Paradigm: Technique-Gated Progression

Dr. Aaron Horschig of Squat University introduces a critical biomechanical constraint to the progressive overload model: movement competency must precede performance capacity.¹⁶ Horschig's central philosophy, "Move first and perform second," suggests that adding weight to a "faulty platform" of poor movement patterns is a primary driver of non-contact injuries, such as ACL ruptures.¹⁶

Biomechanical Markers and Torque Calculations

In the Squat University framework, the decision to increase load is gated by specific technical markers.¹⁶ Biomechanics in the squat are understood through the lens of torque—the force causing rotation around a joint.¹⁹ Torque (T) is defined as the product of force (F) and the moment arm (MA):

$$T = F \times MA$$

The length of the moment arm (the perpendicular distance from the joint to the line of gravity) determines the amount of torque placed on the knee, hip, and low back.¹⁹ For example, a front squat places the longest moment arm on the knee joint, increasing quadriceps demand, while a low-bar back squat increases the moment arm at the hip, allowing for higher mechanical leverage and heavier absolute loads.¹⁹

Squat Variation	Primary Torque Focus	Moment Arm Characteristics
Front Squat	Knee Joint	Longest knee moment arm; upright torso reduces lumbar torque ¹⁹
High-Bar Back Squat	Balanced Knee/Hip	Moderate torque distribution; requires significant ankle mobility ¹⁹
Low-Bar Back Squat	Hip Joint	Maximizes posterior chain leverage; minimizes knee moment arm ¹⁹

Technical Failure and the Squat Screen

Progressive overload should be halted if "technical failure" occurs, even if muscular failure is not reached.¹⁶ Horschig identifies several failure triggers:

- **Knee Valgus:** If the knees collapse inward during the ascent, it indicates a failure of the lateral hip stabilizers to maintain joint alignment under stress.¹⁶
- **Ankle Mobility:** If the user’s heels lift off the ground or the torso collapses forward excessively, it indicates that the current load or depth exceeds their functional mobility.¹⁶
- **Pelvic Control:** The inability to maintain a neutral spine ("butt wink") at the bottom of a

squat suggests that the overload is compromising structural integrity.¹⁸

For an AI workout builder, this implies that the user must "clear" a technique checklist for each session. If the user cannot perform a deep bodyweight squat with perfect alignment, the program should prioritize mobility drills and "Barbell Ready" warm-ups—including barefoot ankle rocks, glute bridges with pauses, and thoracic wrap stretches—before increasing external load.¹⁶

The Shiv Wilson Method: Load Management and Holistic Filters

Sebastian "Shiv" Wilson proposes a hierarchical training model that treats recovery and individualization as the primary decision filters for progression.²¹ Wilson's framework distinguishes between the "7 Principles" (the textbook laws of adaptation) and the "4 Pillars" (the practical execution strategy).¹⁵

The 7 Principles of Strength Training

Wilson utilizes these principles to ensure every repetition is effective and sustainable:

1. **Specificity:** Movement selection must align with the primary goal (e.g., deadlifts for posterior chain strength vs. general fitness).¹⁵
2. **Overload:** Systematically increasing demand to elicit adaptation.¹⁵
3. **Progression:** Ensuring increases are small and repeatable (e.g., +2.5 lbs/week).²¹
4. **Variation:** Changing stimuli to avoid plateaus, though not at the expense of consistency.¹⁵
5. **Individualization:** Adjusting for the user's unique recovery capacity and injury history.¹⁵
6. **Recovery:** Treating rest as a non-negotiable physiological requirement.¹⁵
7. **Reversibility:** Acknowledging that gains are lost if the stimulus is removed (involution).²¹

The 4 Pillars and the 5-Step Audit

To manage these principles daily, Wilson advocates for a "Hybrid Audit Model" that prioritizes recovery over overload.²² The 4 Pillars—Movement Integrity, Progressive Load Management, Consistent Practice, and Integrated Recovery—serve as daily levers.¹⁵

Pillar	AI GPT Programming Logic
Movement Integrity	If technique breaks down, decrease weight or volume immediately ¹⁵

Load Management	Track quantifiable data (reps, weight, rest) to avoid "aspirational" jumps ¹⁵
Consistent Practice	Prioritize 80% adherence to scheduled sessions over 100% intensity in a few ²²
Integrated Recovery	Adjust today's load based on last night's sleep and subjective energy ¹⁵

Wilson’s "5-Step Filter" provides the algorithmic structure for an AI:

- 1. Define the primary goal precisely (e.g., "carry groceries without fatigue").²¹
- 2. Audit recovery capacity (sleep, stress, daily steps).²¹
- 3. Map current progression rhythm (have loads increased in the last 3 weeks?).²¹
- 4. Identify recurring pain points or plateaus.²¹
- 5. Remove "advanced" clutter and rebuild only the necessary variables.²¹

Biological Individuality: Sex-Based Considerations in Training

When designing programs for women, the progressive overload model must account for significant physiological divergences in muscle fiber composition, metabolic efficiency, and recovery kinetics.²⁴

Muscle Fiber Type and Fatigability

Research consistently indicates that women possess a higher proportional area of Type I (slow-twitch) muscle fibers and a lower proportional area of Type II (fast-twitch) fibers compared to men.²⁴ While men are typically capable of generating force more quickly due to larger fiber cross-sectional areas (CSA), women demonstrate superior resistance to muscle fatigue.²⁵

This fatigue resistance means that women can often perform more repetitions at a higher percentage of their one-rep max (1RM) than men.²⁶ In an AI programming context, this suggests that women may benefit from higher relative volume and shorter rest intervals, as they recover more quickly between sets.²⁵ However, it is important to note that while women recover quickly between sets, their recovery of explosive power (e.g., vertical jump) after a high-intensity session may take longer than it does for men.²⁶

Metabolic and Nutritional Nuances

Women are more reliant on aerobic metabolism and fat oxidation during exercise, whereas men utilize more anaerobic pathways.²⁵ This metabolic profile supports higher training volume

tolerance but may necessitate specific nutritional interventions to optimize overload. For instance, women naturally have 70–80% lower intramuscular creatine stores than men and often consume less dietary creatine, making supplementation a powerful "foundational" lever for improving strength, power output, and training volume tolerance.³⁰

Hormonal Periodization: The Menstrual Cycle Variable

The menstrual cycle introduces a dynamic macro-variable into the progressive overload model. Fluctuations in estrogen and progesterone significantly impact energy levels, recovery speed, and strength potential.³¹

Phase 1: Menstrual (Days 1–5)

During this phase, estrogen and progesterone are at their lowest.³¹ While many women experience fatigue or cramps, the lower core body temperature can support endurance training.³¹ The AI should prioritize gentle movement (yoga, walking, Zone 2 cardio) early in the period, gradually returning to strength work as estrogen begins to rise toward the end of the phase.³¹

Phase 2: Follicular (Days 6–12)

Estrogen rises gradually, peaking just before ovulation.³¹ Estrogen is highly anabolic; it supports muscle protein synthesis, enhances glucose uptake, and acts as an anti-inflammatory, reducing exercise-induced muscle damage.²⁵ This is the optimal window for heavy lifting, high-intensity interval training (HIIT), and PR (personal record) attempts.³¹ The neuromuscular system is "primed" for skill acquisition and motor learning during this peak estrogen window.³¹

Phase 3: Ovulatory (Days 13–15)

Strength and power output often peak during ovulation due to maximal estrogenic support.³¹ However, increased estrogen can lead to greater ligament laxity, slightly increasing the risk of injury.³² Programming should focus on high intensity but with a strict emphasis on warm-ups and technical integrity to mitigate this vulnerability.³¹

Phase 4: Luteal (Days 16–28)

Progesterone peaks during the mid-luteal phase, while estrogen declines.³¹ Progesterone is catabolic and increases core body temperature, which can lead to faster fatigue, reduced heat tolerance, and slower recovery.³¹ Many women experience higher perceived exertion and reduced motivation.³² The AI should shift the focus to moderate-intensity steady-state cardio, maintenance-level strength training, and increased recovery strategies, such as deloading or lower-impact conditioning.³¹

Cycle Phase	Hormonal Profile	Training Recommendation	Recovery Strategy
Menstrual	Low Estrogen/Progesterone	Mobility, Zone 2, gentle strength ³¹	Iron-rich foods, restorative rest ³³
Follicular	Rising Estrogen	Heavy loads, HIIT, PR attempts ³¹	High energy availability; muscle growth focus ³¹
Ovulatory	Peak Estrogen	Maximal strength, explosive work ³²	Strict technique focus; thorough warm-up ³¹
Luteal	High Progesterone	Moderate intensity, technique work ³¹	Deloading, increased sleep, hydration ³¹

Pathophysiology of Excessive Load: Overtraining Markers

To prevent the transition from functional overreaching to overtraining syndrome (OTS), the programming logic must incorporate "if-then" constraints based on the 19 markers of overtraining identified by the National Academy of Sports Medicine (NASM).³⁵ These markers are categorized into exercise-related symptoms, physical/mental changes, and systemic health concerns.

Exercise-Related and Physical Symptoms

A primary indicator of overtraining is a performance plateau or decline despite continued effort.¹¹ Users may report an increased perceived exertion during "normal" weights, excessive sweating, or unusual feelings of heaviness and stiffness.³⁵ If a user reports "not feeling refreshed" after a rest day, it signals that the fatigue accumulation is outpacing the recovery cycle (SRA).¹¹

Systemic and Psychological Concerns

Overtraining manifests systemically through an increase in resting heart rate (RHR) or resting blood pressure, loss of appetite, and chronic joint pain.¹⁰ Psychological markers include a decline in motivation, mood disturbances (agitation, irritability), and new-onset insomnia or

poor sleep quality.¹⁰ For female trainees, changes or cessation of menstruation (amenorrhea) is a critical systemic marker that the current training load and nutritional intake are insufficient to maintain hormonal health.³⁵

Biochemical Mechanisms of Fatigue

The "Central Fatigue Hypothesis" suggests that overtraining is linked to a disruption in the neurotransmitter serotonin (5-HT).¹⁰ Intense exercise increases levels of unbound tryptophan in the brain, which is the precursor to serotonin.¹⁰ Simultaneously, exercise-induced oxidation of branched-chain amino acids (BCAAs) reduces the competition for tryptophan to enter the brain, leading to increased 5-HT synthesis and persistent feelings of fatigue and lethargy.¹⁰ While supplementation with BCAAs can sometimes dampen this effect, the primary solution is a significant reduction in training volume.¹⁰

Periodization Models and Progression Architectures

To manage the long-term application of progressive overload, programs utilize periodization models that cycle training variables over weeks (microcycles), months (mesocycles), and years (macrocycles).⁴

Linear Periodization (LP)

The traditional linear model is characterized by a high initial volume and low intensity, transitioning gradually toward low volume and high intensity.³⁶ This model is ideal for beginners because it provides a predictable and steady sequence of loading that removes "guesswork".³⁸ It allows for a solid foundation of muscle size and connective tissue strength to be built before exposing the joints to maximal loads.¹²

Undulating Periodization (UP)

Undulating models vary intensity and volume more frequently, either workout-to-workout (Daily Undulating Periodization - DUP) or week-to-week (Weekly Undulating Periodization - WUP).³⁶ This frequent variation helps trained lifters avoid the "repeated bout effect," where the body becomes too efficient at a single stimulus and ceases to adapt.³⁹ Research suggests that for trained individuals, undulating models can lead to larger strength gains than linear models.³⁷

Flexible Non-Linear (FNL) Programming

The most sophisticated approach for a custom AI would be a Flexible Non-Linear model, which adjusts variables based on the user's physiological and psychological readiness.³⁶ This "auto-regulation" allows the AI to pivot the program in real-time. For instance, if a user logs poor sleep and high stress, the FNL model might swap a "Max Strength" day for a "Mobility/Technique" day, ensuring the overload is always productive and never destructive.¹⁵

Synthesizing the Data: Algorithmic "If-Then" Programming Logic

To translate this research into a functional AI GPT for building workout programs, the following logic gates must be implemented. These rules synthesize the philosophies of Squat University (technique-gating), Shiv Wilson (recovery-filtering), and sex-specific hormonal realities.

Daily Pre-Workout Readiness Filters

1. **If** Sleep Duration < 6 hours **OR** Subjective Energy $< 2/5$, **then** reduce today's target sets by 50% and intensity by 10%.²²
2. **If** Resting Heart Rate is > 5 bpm above baseline, **then** prioritize active recovery (Zone 2 cardio, stretching) and skip heavy lifting.³⁵
3. **If** Joint Pain is reported as "Sharp" or "Radiating," **then** halt the exercise immediately and swap for a non-painful regression (e.g., swap Barbell Squat for Goblet Squat).¹⁵

Weekly Progression Logic Gates

1. **If** all sets were completed with "Perfect Technique" (no knee valgus, neutral spine) **AND** Reps in Reserve (RIR) was ≥ 2 , **then** increase load by 2.5–5 lbs or 1–2 reps for the next session.²
2. **If** "Technical Failure" occurred during any set (e.g., knees caved in, heels lifted), **then** maintain current load and add one specific "Barbell Ready" mobility drill to the warm-up for that movement pattern.¹⁶
3. **If** a primary lift has not increased in load or reps for 3 consecutive weeks, **then** introduce a "Variation" overload (e.g., add a 3-second eccentric or a pause at the bottom) rather than adding weight.²¹

Menstrual Cycle Programming Adjustments

1. **If** Days 6–14 (Follicular/Ovulatory), **then** prioritize "Overload" through Absolute Load and HIIT; this is the window for PR attempts.³¹
2. **If** Days 21–28 (Late Luteal), **then** prioritize "Recovery" and "Density"; reduce absolute weight and sets, focus on steady-state movement and technique maintenance.³¹
3. **If** significant PMS symptoms are reported (cramps, severe bloating), **then** trigger a "Deload Week" regardless of the current phase in the training block.³¹

Overtraining "Hard Reset" Rules

1. **If** the user exhibits ≥ 3 markers of overtraining (e.g., insomnia, performance drop, persistent soreness), **then** mandate 7 days of "Offloading" (total rest or walking only).³⁵

2. If performance does not return to baseline after a 1-week deload, **then** "Redirect" the training to a different modality (e.g., swap barbell work for swimming or yoga) for an additional 7–14 days.³⁵

Strategic Implementation for Automated Programs

The final integration of this model requires a weekly structure that balances these competing needs. For a female trainee, the most effective "Progressive Overload" is one that is rhythmically aligned with her biology. By treating "Movement Integrity" as the prerequisite and "Recovery" as the ultimate decision-maker, the AI can ensure that every increment of load contributes to long-term strength and capability. This model moves away from the "more is always better" mentality and toward a sophisticated management of biological stress, ensuring the user is bulletproofed from injury while making measurable, sustainable strides toward their athletic potential.¹⁵

Works cited

1. accessed March 2, 2026, <https://www.muscleandstrength.com/articles/progressive-overload-the-complete-guide#:~:text=Progressive%20overload%20is%20a%20training.%2C%20muscle%20growth%2C%20and%20performance.>
2. Progressive Overload 101: What It Is, How It Works, and How to Apply It | Muscle & Strength, accessed March 2, 2026, <https://www.muscleandstrength.com/articles/progressive-overload-the-complete-guide>
3. Why Progressive Overload is Key to Getting Real Results - Tonal, accessed March 2, 2026, <https://tonal.com/blogs/all/progressive-overload>
4. Progressive Overload Explained: Grow Muscle & Strength Today - NASM Blog, accessed March 2, 2026, <https://blog.nasm.org/progressive-overload-explained>
5. The Science of Supercompensation and How It Makes You Fast - TrainerRoad, accessed March 2, 2026, <https://www.trainerroad.com/blog/science-of-supercompensation/>
6. Understand the General Adaptation Syndrome Model - Human Kinetics, accessed March 2, 2026, <https://us.humankinetics.com/blogs/excerpt/understand-the-general-adaptation-syndrome-model>
7. How Cycling Gains are Made | Highland Training, accessed March 2, 2026, <https://www.highlandtraining.net/building-cycling-strength-how-it-happens>
8. Central Concepts Related to Periodization - NSCA, accessed March 2, 2026, <https://www.nsc.com/education/articles/kinetic-select/central-concepts-related-to-periodization/>
9. Supercompensation Time Course or Mechanism? - EliteFTS, accessed March 2, 2026, <https://elitefts.com/blogs/coaching-logs/supercompensation-time-course-or-me>

[chanism](#)

10. Overtraining Syndrome: A Practical Guide - PMC, accessed March 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3435910/>
11. What is Overtraining? - The National Sports Medicine Institute, accessed March 2, 2026, <https://www.nationalsportsmed.com/what-is-overtraining/>
12. Progressive Overload: The Strength Principle You Need to Know | Anytime Fitness, accessed March 2, 2026, <https://www.anytimefitness.com/blog/progressive-overload>
13. Progressive Overload: The Secret to Building Muscle Strength - University Hospitals, accessed March 2, 2026, <https://www.uhhospitals.org/blog/articles/2025/08/progressive-overload>
14. Lean mass sparing in resistance-trained athletes during caloric restriction - PMC - NIH, accessed March 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9012799/>
15. How to Apply the 4 Principles of Strength Training Effectively - Wellness Hub, accessed March 2, 2026, <https://wellness.alibaba.com/fitlife/4-principles-of-strength-training>
16. The Movement Looking Glass - Squat University, accessed March 2, 2026, <https://squatuniversity.com/2015/10/15/the-movement-looking-glass/>
17. Squat University - The Ultimate Guide To Finding Your True Strength, accessed March 2, 2026, <https://squatuniversity.com/>
18. Biomechanics - Squat University, accessed March 2, 2026, <https://squatuniversity.com/category/biomechanics/>
19. The Real Science of the Squat - Squat University, accessed March 2, 2026, <https://squatuniversity.com/2016/04/20/the-real-science-of-the-squat/>
20. Squat U Club: Training by Dr. Aaron Horschig in TrainHeroic, accessed March 2, 2026, <https://marketplace.trainheroic.com/workout-plan/team/squat-u-club>
21. The 7 Principles of Strength Training Guide - Wellness Hub, accessed March 2, 2026, <https://wellness.alibaba.com/fitlife/7-principles-of-strength-training>
22. How to Apply the 4 Principles and Pillars of Lifting Guide, accessed March 2, 2026, <https://wellness.alibaba.com/fitlife/4-principles-pillars-of-strength-training>
23. Qualitative Comparison of Periodisation Principles and Methodology - Qwik Kiwi Coaching, accessed March 2, 2026, <https://www.coachray.nz/2022/02/28/qualitative-comparison-of-periodisation-principles-and-methodology/>
24. Sex differences in human skeletal muscle fiber types and the influence of age, physical activity, and muscle group: A systematic review and meta-analysis - PMC, accessed March 2, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12580412/>
25. Gender Differences in Training - The Muscle PhD, accessed March 2, 2026, <https://themusclephd.com/gender-differences-in-training/>
26. Sex Differences in the Temporal Recovery of ... - Frontiers, accessed March 2, 2026, https://www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2018.01480/full?utm_source=chatgpt.com
27. Sex differences in absolute and relative changes in muscle size following

- resistance training in healthy adults: a systematic review with Bayesian meta-analysis - PMC, accessed March 2, 2026,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11869894/>
28. Sex differences in skeletal muscle fiber types: A meta-analysis - Edith Cowan University, accessed March 2, 2026,
https://ro.ecu.edu.au/context/ecuworks2022-2026/article/3809/viewcontent/Sex_20differences_20in_20skeletal_20muscle_20fiber_20types.pdf
 29. volume recommendations for tall lifters? - The Barbell Medicine Forum, accessed March 2, 2026,
<https://forum.barbellmedicine.com/t/volume-recommendations-for-tall-lifters/8929>
 30. The Creatine Guide for Active Women - Dr Stacy Sims, accessed March 2, 2026,
<https://www.drstacysims.com/newsletters/articles/posts/creatine-guide-for-active-women>
 31. Optimize Your Workouts to Your Menstrual Cycle - Mend Colorado, accessed March 2, 2026,
<https://www.mendcolorado.com/physical-therapy-blog/2025/12/18/optimize-your-workouts-to-your-menstrual-cycle/>
 32. Exercise and Training During the Menstrual Cycle - Nashville Physical Therapy, accessed March 2, 2026,
<https://www.nashvillephysicaltherapy.com/post/exercise-and-training-during-the-menstrual-cycle-how-to-optimize-your-workouts-based-on-your-hormon>
 33. Hormone Cycle, Training and Recovery Tips for Female Athletes - USA Physique, accessed March 2, 2026,
<https://www.teamusaphysique.com/post/hormone-cycle-training-and-recovery-tips-for-female-athletes>
 34. Optimizing Training for Females - Fit For Life, accessed March 2, 2026,
<https://www.fitforlifephysicaltherapy.com/new-page-75>
 35. 19 Signs of Overtraining: How to Avoid Excess Fatigue and OTS, accessed March 2, 2026, <https://blog.nasm.org/strategies-for-overcoming-overtraining>
 36. Muscle Daily Undulating Periodization for Strength and Body Composition: The Proposal of a New Model - PMC, accessed March 2, 2026,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9987427/>
 37. Comparing Linear and Undulating Periodization - University Digital Conservancy, accessed March 2, 2026,
<https://conservancy.umn.edu/bitstreams/c26dd647-2648-4003-9550-00afc2a902e3/download>
 38. CURRENT CONCEPTS IN PERIODIZATION OF STRENGTH AND CONDITIONING FOR THE SPORTS PHYSICAL THERAPIST - PMC, accessed March 2, 2026,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4637911/>
 39. Undulating Periodization: Definition, Types, Uses, and Examples - Hevy Coach, accessed March 2, 2026,
<https://hevycoach.com/glossary/undulating-periodization/>
 40. A Comprehensive Overview of the Periodization Research : r/Fitness - Reddit, accessed March 2, 2026,

https://www.reddit.com/r/Fitness/comments/7p27br/a_comprehensive_overview_of_the_periodization/

41. Certified™: April 2025 - Menstrual Cycle and Resistance Training: What the Latest Research Says - ACE, accessed March 2, 2026,
<https://www.acefitness.org/continuing-education/certified/april-2025/8846/menstrual-cycle-and-resistance-training-what-the-latest-research-says/>